

HARSHITH DESHALLI RAVI

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EDUCATION

Illinois Institute of Technology, M. S. Data Science

Aug 2023 - Dec 2025

Selective Coursework: Advanced Machine Learning, Deep Learning for Robotics, Computer Vision, Reinforcement Learning, Robotics Simulation, Edge AI, Motion Planning, Probabilistic Robots, Advanced Statistics, Monte-Carlo Statistics, Generative AI Models.

Global Academy of Technology, B.E. Electrical and Electronics

Aug 2019 - Jul 2023

Selective Coursework: Robotics Systems and Control, Modern Control Theory, Sensors and Actuators, Embedded Systems Design, Motor Design, Digital Signal Processing (DSP), Linear Algebra, Advanced Calculus, Battery Management System (BMS), Computer Architecture.

EXPERIENCE

Robotics Researcher

Illinois Institute of Technology, USA

Sep 2024 – Present

- **Enhanced operator control accuracy by 40%** and achieved **<100 ms force-feedback latency with 1 mm positional precision** through optimized real-time control loops and calibrated feedback motors.
- **Constructed a 5-DOF wearable exoskeleton** enabling **real-time haptic teleoperation**, providing tactile feedback of remote robot forces for intuitive and responsive control.
- **Integrated ROS, motor control systems, and VR hardware** within **NVIDIA Isaac Sim** to synchronize user body motion with robot movement, enabling a complete **sim-to-real digital twin** workflow.
- **Validated through 20+ simulation and real-world trials**, demonstrating significant improvement in immersion, stability, and system responsiveness during teleoperation tasks.

Robotics Research Assistant

Indian Institute of Technology Bengaluru, India

Oct 2022 – Apr 2023

- **Improved learning rate by 50%** among **30+ children with autism** by deploying interactive social robots that taught balance, path following, and language using **emotion-aware cues** and **emoji-based expressions**.
- **Designed and programmed two mobile robots** powered by **ROS2, Raspberry Pi, dual cameras**, and **ultrasonic sensors** for adaptive movement and behavior-based engagement.
- **Achieved ~95% tracking accuracy** in movement through **PID control** and **path-planning algorithms**, enabling precise robot motion in dynamic classroom environments.
- **Enhanced robot response and stability by 30%** via improved hardware–software integration, sensor calibration, and optimized control loops for smooth human–robot interaction

Edge AI & Computer Vision Engineer

DHI Flagship and Innovation Centre, India

Jun 2021 – Nov 2022

- Achieved **95% object detection accuracy** across a **50,000+ image dataset** by deploying real-time detection models for **public transport and currency recognition** using **Python, OpenCV, and YOLOv5**.
- Reduced **inference latency by 30%** on **Raspberry Pi and Jetson** devices through model compression and **TensorFlow Lite conversion**, maintaining accuracy above 95% for edge deployment.
- **Enhanced accessibility and response time by 30%** for visually impaired users by integrating an **NLP-driven speech interface** built with **NLTK, Apache Spark, and SQL**, containerized with **Docker** for scalable rollout.

SKILLS

Programming & Software: Python, C++, ROS/ROS2, Git, Real-Time Microcontroller Programming, CI/CD Pipelines, Docker.

Robotics & AI: Robot Kinematics & Dynamics, Motion & Trajectory Planning (RRT*, PID, MPC), SLAM & Sensor Fusion, ML & DL (PyTorch, TensorFlow, JAX), Reinforcement Learning (DQN, PPO), Computer Vision (OpenCV, YOLO, 3D Point Clouds).

Simulation & Hardware: Gazebo, Isaac Sim, Digital Twin Simulation, Sensor Integration (IMU, LiDAR, Cameras), Embedded Systems (Arduino, STM32, Raspberry Pi), CAD/CAM (SolidWorks, Fusion 360)

ACADEMIC PROJECTS

3D LiDAR-Based SLAM and Autonomous Navigation using ROS2 and Isaac Sim

- Revamped **fully autonomous navigation** with accurate localization and smooth trajectory planning in a simulated environment (**NVIDIA Omniverse**) using **Nav2** and real-time **collision avoidance**.
- Generated high-fidelity **3D point cloud maps** using **Cartographer** and **3D LiDAR**, converted them into **2D costmaps**, and integrated via an **Isaac Sim–ROS2 bridge** for seamless sensor data exchange and control.

AI Manipulator: Multi-Modal Robotic Control using Isaac Sim and NVIDIA Groot N1.5

- Achieved **real-time multi-modal manipulation** (gesture, audio, and perception-driven control) on the **Open Manipulator X** by fine-tuning **NVIDIA's Groot N1.5 foundation model** using large-scale synthetic data from a custom **Isaac Sim** environment.
- Integrated **ROS2-based simulation–hardware pipeline** with a custom AI interface on **Jetson Nano** (dual cameras + depth sensor), enabling **real-time inference, motion planning, and adaptive pick-and-place execution**.

ACHIEVEMENTS

- Secured **Top 5 global ranking** out of 4,500+ participants at **TikTok TechJam**, with **5+ podium finishes** in **10+ national and international hackathons**
- **Led the design and development** of an **electric go-kart**, achieving a **Top 5 finish** at the **Indian National Kart Racing (IKR) Championship**, representing the state team.